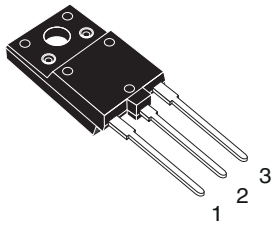
MD2310FX

High voltage NPN Power transistor for standard definition CRT display

Features

- State-of-the-art technology:
  - diffused collector “enhanced generation”
- More stable performance versus operating temperature variation
- Low base drive requirement
- Tighter h_{FE} range at operating collector current
- Fully insulated power package u.l. compliant
- In compliance with the 2002/93/EC european directive

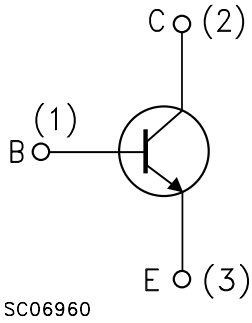


ISOWATT218FX

Applications

- Horizontal deflection output for monitor and real flat tv

Internal Schematic Diagram



Description

The MD2310FX is manufactured using Diffused Collector in Planar Technology adopting new and enhanced high voltage structure. The new MD product series show improved silicon efficiency bringing updated performance to the Horizontal Deflection stage.

Order codes

Part number	Marking	Package	Packing
MD2310FX	MD2310FX	ISOWATT218FX	TUBE

1 Electrical ratings

Table 1. Absolute maximum rating

Symbol	Parameter	Value	Unit
V_{CES}	Collector-emitter voltage (V_{BE} = 0)	1500	V
V_{CEO}	Collector-emitter voltage (I_B = 0)	700	V
V_{EBO}	Collector-base voltage (I_C = 0)	9	V
I_C	Collector current	14	A
I_{CM}	Collector peak current (t_P < 5ms)	21	A
I_B	Base current	7	A
P_{TOT}	Total dissipation at T_c = 25°C	62	W
V_{inso}	Insulation withstand voltage (RMS) from all three leads to external heatsink	2500	V
T_{stg}	Storage temperature	-65 to 150	°C
T_J	Max. operating junction temperature	150	

Table 2. Thermal data

Symbol	Parameter		Value	Unit
$R_{thj\text{-}case}$	Thermal resistance junction-case	Max	2	°C/W

2 Electrical characteristics

(T_{CASE} = 25°C; unless otherwise specified)

Table 3. Electrical characteristics

Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Unit
I_{CES}	Collector cut-off current (V_{BE} = 0)	V_{CE} = 1500V V_{CE} = 1500V T_c= 125°C			0.2 2	mA mA
I_{EBO}	Emitter cut-off current (I_C = 0)	V_{EB} = 9V			1	mA
$V_{CEO(sus)}$	Collector-emitter sustaining voltage (I_C = 0)	I_C = 100 mA	700			V
$V_{CE(sat)}$ *Note 1*	Collector-emitter saturation voltage	I_C = 7 A I_B = 1.75 A			2.5	V
$V_{BE(sat)}$ *Note 1*	Base-emitter saturation voltage	I_C = 7 A I_B = 1.75 A			1.1	V
h_{FE} *Note 1*	DC current gain	I_C = 1 A V_{CE} = 5 V I_C = 7 A V_{CE} = 1 V I_C = 7 A V_{CE} = 5 V	 6	28 5.5 	 8.5	
 t_s t_f	INDUCTIVE LOAD Storage time Fall time	I_C = 6A f_h = 64KHz $I_{B(on)}$ = 0.9A $V_{BE(off)}$ = -2.7V $L_{BB(off)}$ = 1.6µH		 2.3 0.12	 2.8 0.25	 µs µs

1 Pulsed duration = 300 µs, duty cycle ≤1.5%.

2.1 Typical characteristics

Figure 1. Safe operating area

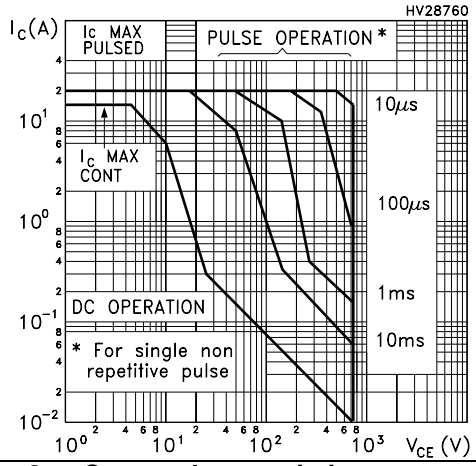


Figure 2. Derating curve

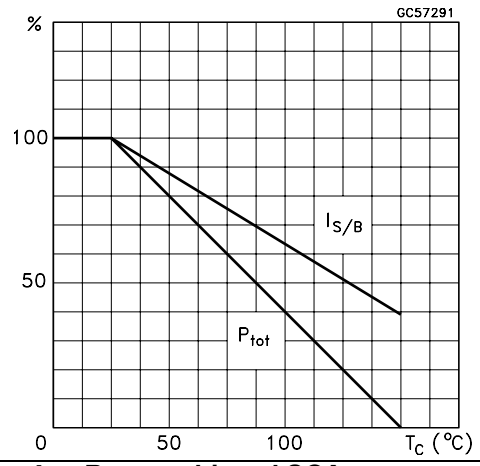


Figure 3. Output characteristics

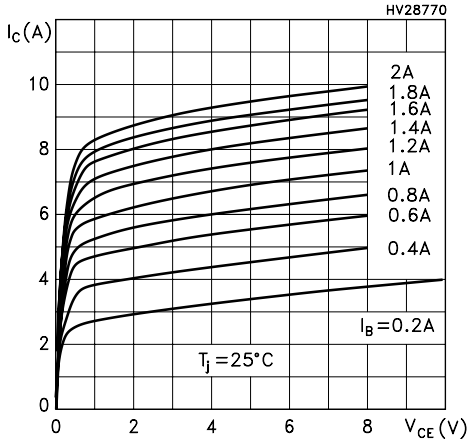


Figure 4. Reverse biased SOA

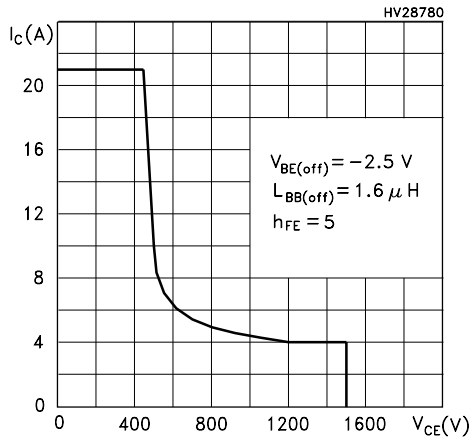


Figure 5. DC current gain

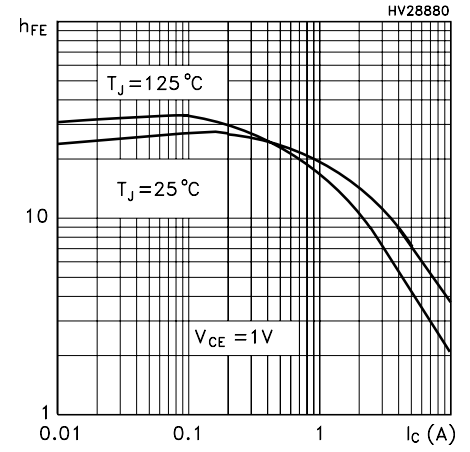


Figure 6. DC current gain

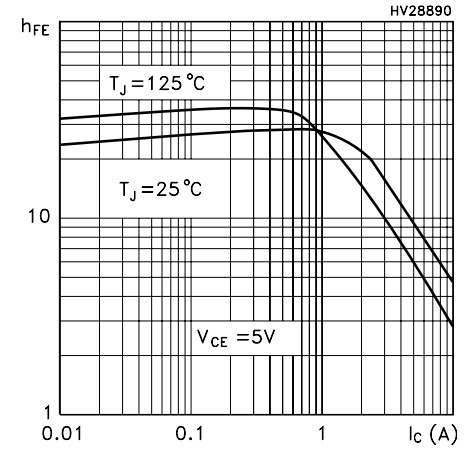


Figure 7. Collector-emitter saturation voltage

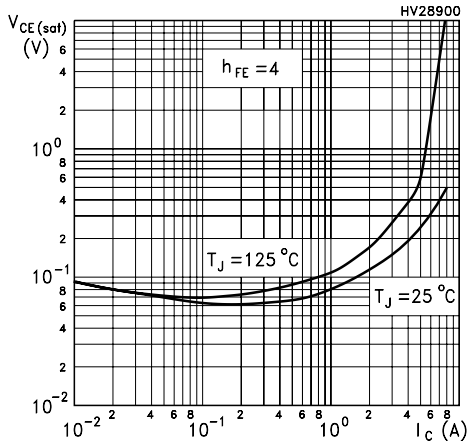


Figure 8. Base-emitter saturation voltage

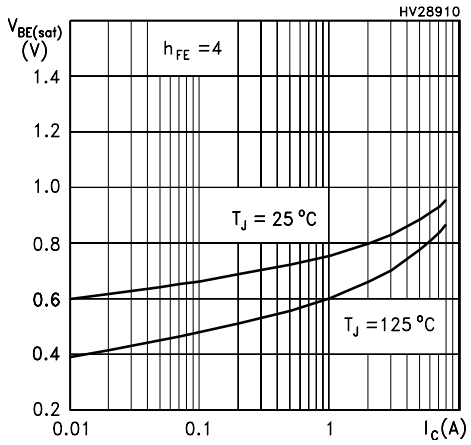


Figure 9. Power losses

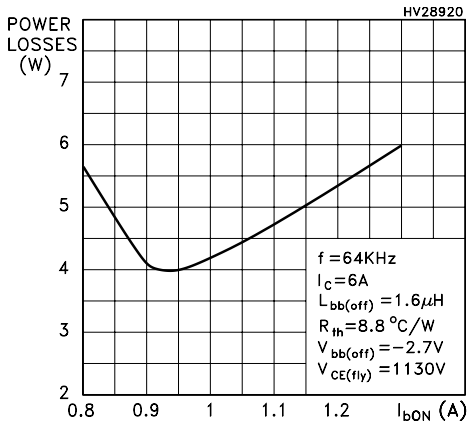
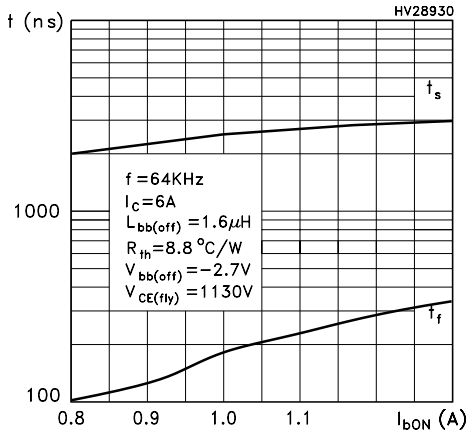


Figure 10. Inductive load switching time



3 Test circuits

Figure 11. Power losses and inductive load switching test circuit

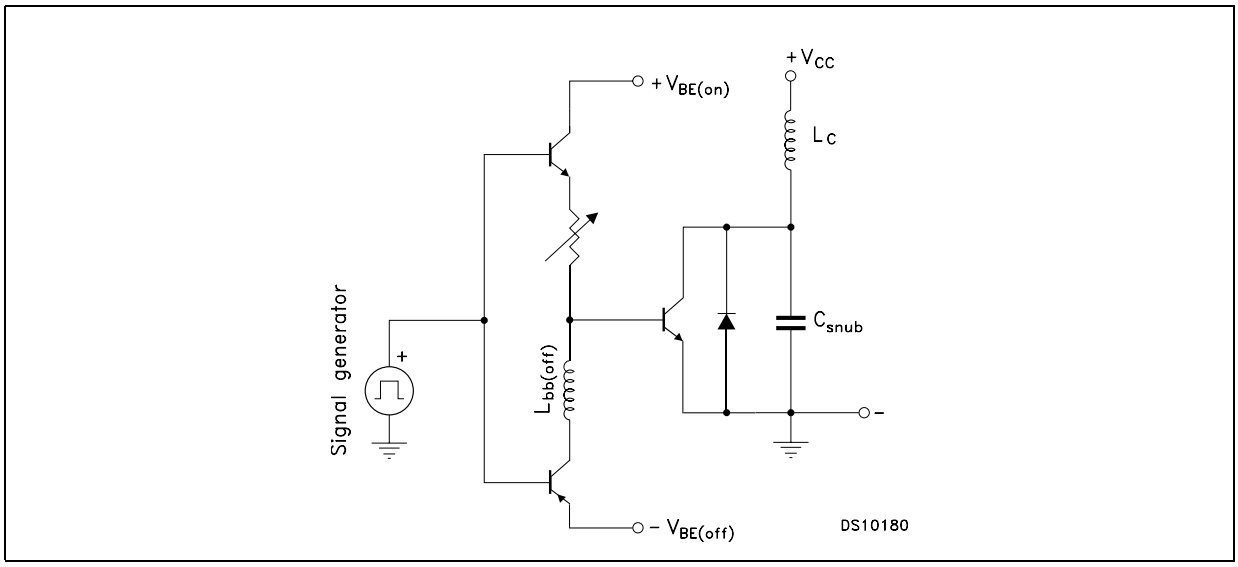
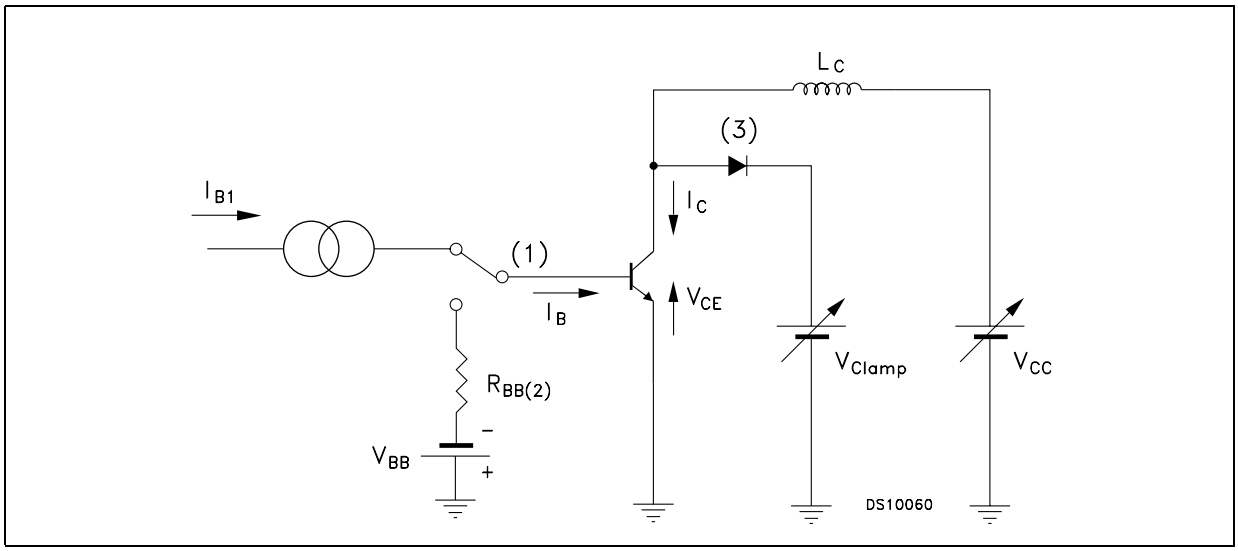


Figure 12. Reverse biased safe operating area test circuit

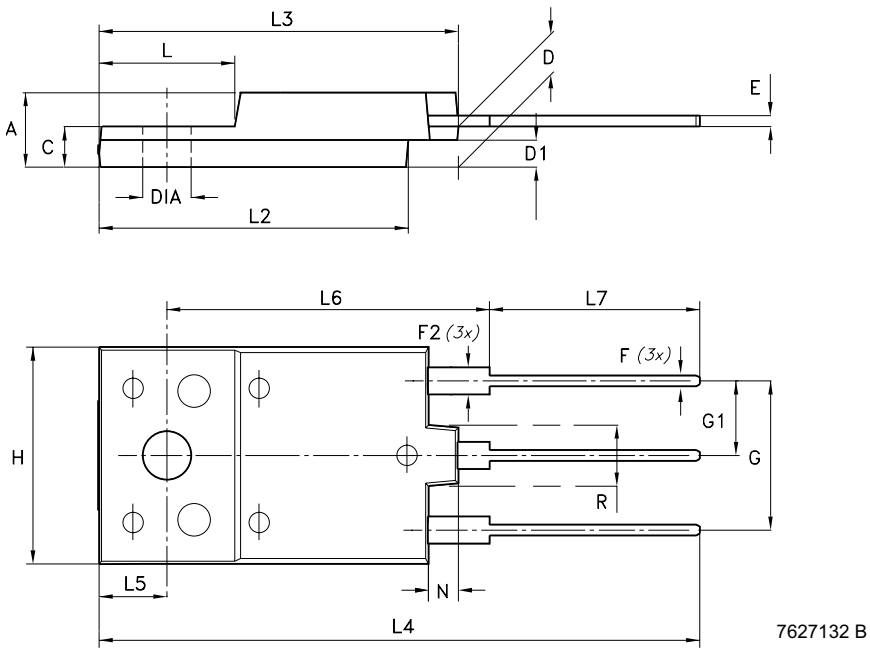


4 Package mechanical data

In order to meet environmental requirements, ST offers these devices in ECOPACK® packages. These packages have a Lead-free second level interconnect . The category of second level interconnect is marked on the package and on the inner box label, in compliance with JEDEC Standard JESD97. The maximum ratings related to soldering conditions are also marked on the inner box label. ECOPACK is an ST trademark. ECOPACK specifications are available at: www.st.com

ISOWATT218FX MECHANICAL DATA

DIM.	mm.		
	MIN.	TYP	MAX.
A	5.30		5.70
C	2.80		3.20
D	3.10		3.50
D1	1.80		2.20
E	0.80		1.10
F	0.65		0.95
F2	1.80		2.20
G	10.30		11.50
G1		5.45	
H	15.30		15.70
L	9		10.20
L2	22.80		23.20
L3	26.30		26.70
L4	43.20		44.40
L5	4.30		4.70
L6	24.30		24.70
L7	14.60		15
N	1.80		2.20
R	3.80		4.20
Dia	3.40		3.80



5 Revision history

Table 4. Revision history

Date	Revision	Changes
18-Oct-2005	1	First release
25-Nov-2005	2	Complete datasheet
15-Dec-2005	3	Legal page inserted
29-Sep-2006	4	New h_{FE} limit

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